Class 11: Protein Structure Prediction

Your Name

Setup

```
library(bio3d)
library(pheatmap)
library(jsonlite)

setwd("/Users/omar/BIMM143/Lab 11")
```

Loading AlphaFold Results

```
results_dir <- "/Users/omar/BIMM143/Lab 11/hivprdimer_23119/"

pdb_files <- list.files(path=results_dir,
                        pattern="*.pdb",
                        full.names = TRUE)

basename(pdb_files)
```

```
[1] "hivprdimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb"
[2] "hivprdimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb"
[3] "hivprdimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb"
[4] "hivprdimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb"
[5] "hivprdimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb"
```

Aligning and Superposing Structures

```r
pdbs <- pdbaln(pdb_files, fit=TRUE, exefile="msa")
```

```
Reading PDB files:
/Users/omar/BIMM143/Lab 11/hivprdimer_23119//hivprdimer_23119_unrelaxed_rank_001_alphafold2_r
/Users/omar/BIMM143/Lab 11/hivprdimer_23119//hivprdimer_23119_unrelaxed_rank_002_alphafold2_r
/Users/omar/BIMM143/Lab 11/hivprdimer_23119//hivprdimer_23119_unrelaxed_rank_003_alphafold2_r
/Users/omar/BIMM143/Lab 11/hivprdimer_23119//hivprdimer_23119_unrelaxed_rank_004_alphafold2_r
/Users/omar/BIMM143/Lab 11/hivprdimer_23119//hivprdimer_23119_unrelaxed_rank_005_alphafold2_r
.....

Extracting sequences

pdb/seq: 1   name: /Users/omar/BIMM143/Lab 11/hivprdimer_23119//hivprdimer_23119_unrelaxed_ra
pdb/seq: 2   name: /Users/omar/BIMM143/Lab 11/hivprdimer_23119//hivprdimer_23119_unrelaxed_ra
pdb/seq: 3   name: /Users/omar/BIMM143/Lab 11/hivprdimer_23119//hivprdimer_23119_unrelaxed_ra
pdb/seq: 4   name: /Users/omar/BIMM143/Lab 11/hivprdimer_23119//hivprdimer_23119_unrelaxed_ra
pdb/seq: 5   name: /Users/omar/BIMM143/Lab 11/hivprdimer_23119//hivprdimer_23119_unrelaxed_ra
```

```r
pdbs
```

```
                               1        .         .         .         .         50 
[Truncated_Name:1]hivprdimer   PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:2]hivprdimer   PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:3]hivprdimer   PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:4]hivprdimer   PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:5]hivprdimer   PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
                               ************************************************** 
                               1        .         .         .         .         50 

                              51        .         .         .         .         100 
[Truncated_Name:1]hivprdimer   GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:2]hivprdimer   GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:3]hivprdimer   GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:4]hivprdimer   GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:5]hivprdimer   GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
                               ************************************************** 
                              51        .         .         .         .         100 

                             101        .         .         .         .         150 
[Truncated_Name:1]hivprdimer   QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
```

```
[Truncated_Name:2]hivprdimer   QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:3]hivprdimer   QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:4]hivprdimer   QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:5]hivprdimer   QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
                               **************************************************
                             101         .         .         .         .         150

                             151         .         .         .         .       198
[Truncated_Name:1]hivprdimer   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:2]hivprdimer   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:3]hivprdimer   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:4]hivprdimer   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:5]hivprdimer   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
                               ************************************************
                             151         .         .         .         .       198

Call:
  pdbaln(files = pdb_files, fit = TRUE, exefile = "msa")

Class:
  pdbs, fasta

Alignment dimensions:
  5 sequence rows; 198 position columns (198 non-gap, 0 gap)

+ attr: xyz, resno, b, chain, id, ali, resid, sse, call
```

RMSD Analysis

We can use RMSD to measure structural distance between all pairs of models.

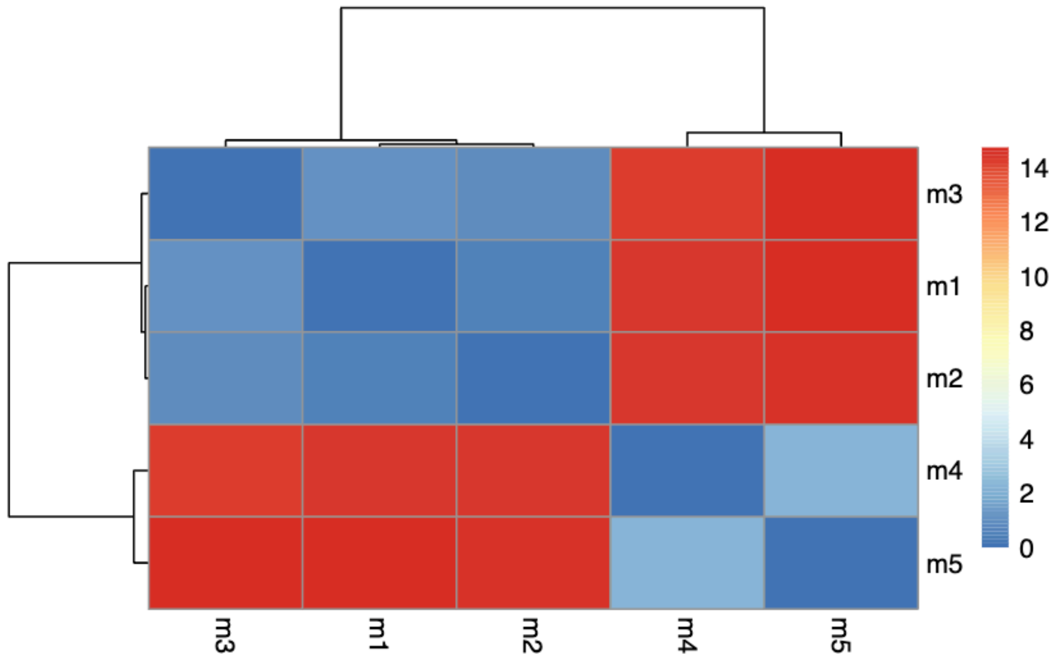
```r
rd <- rmsd(pdbs, fit=T)
```

```
Warning in rmsd(pdbs, fit = T): No indices provided, using the 198 non NA positions
```

```r
range(rd)
```

```
[1]  0.000 14.754
```

```r
colnames(rd) <- paste0("m",1:5)
rownames(rd) <- paste0("m",1:5)
pheatmap(rd)
```



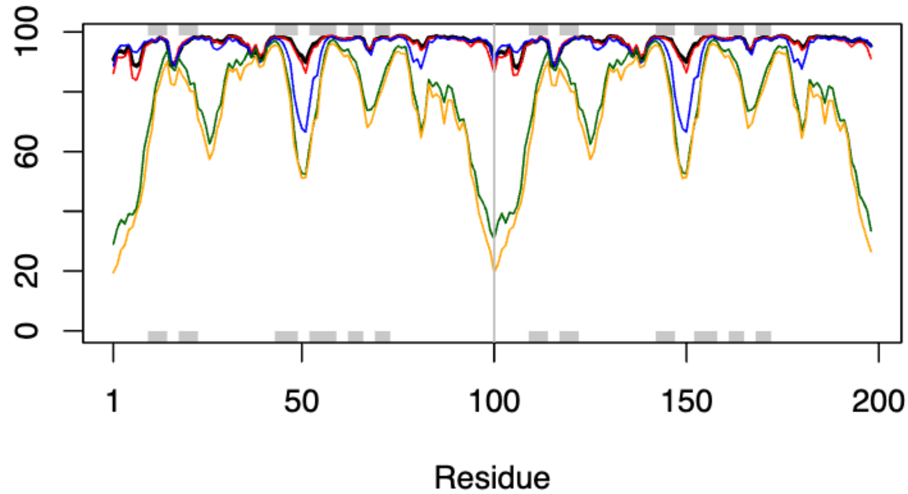
Plotting pLDDT Scores

The pLDDT score is a per-residue confidence metric ranging from 0-100. Values above 70 are considered high confidence.

```r
pdb <- read.pdb("1hsg")
```

```
  Note: Accessing on-line PDB file
```

```r
plotb3(pdbs$b[1,], typ="l", lwd=2, sse=pdb)
points(pdbs$b[2,], typ="l", col="red")
points(pdbs$b[3,], typ="l", col="blue")
points(pdbs$b[4,], typ="l", col="darkgreen")
points(pdbs$b[5,], typ="l", col="orange")
abline(v=100, col="gray")
```



Core Structure Analysis

```r
core <- core.find(pdbs)
```

```
 core size 197 of 198  vol = 9886.027
 core size 196 of 198  vol = 6910.101
 core size 195 of 198  vol = 1339.391
 core size 194 of 198  vol = 1040.703
 core size 193 of 198  vol = 951.872
 core size 192 of 198  vol = 899.097
 core size 191 of 198  vol = 834.744
 core size 190 of 198  vol = 771.349
 core size 189 of 198  vol = 733.076
 core size 188 of 198  vol = 697.291
 core size 187 of 198  vol = 659.753
 core size 186 of 198  vol = 625.285
 core size 185 of 198  vol = 589.553
 core size 184 of 198  vol = 568.266
 core size 183 of 198  vol = 545.027
 core size 182 of 198  vol = 512.902
 core size 181 of 198  vol = 490.736
```

```
core size 180 of 198  vol = 470.28
core size 179 of 198  vol = 450.744
core size 178 of 198  vol = 434.749
core size 177 of 198  vol = 420.351
core size 176 of 198  vol = 406.672
core size 175 of 198  vol = 393.347
core size 174 of 198  vol = 382.408
core size 173 of 198  vol = 372.872
core size 172 of 198  vol = 357.007
core size 171 of 198  vol = 346.581
core size 170 of 198  vol = 337.46
core size 169 of 198  vol = 326.673
core size 168 of 198  vol = 314.965
core size 167 of 198  vol = 304.142
core size 166 of 198  vol = 294.567
core size 165 of 198  vol = 285.663
core size 164 of 198  vol = 278.9
core size 163 of 198  vol = 266.78
core size 162 of 198  vol = 259.01
core size 161 of 198  vol = 247.737
core size 160 of 198  vol = 239.855
core size 159 of 198  vol = 234.978
core size 158 of 198  vol = 230.077
core size 157 of 198  vol = 222
core size 156 of 198  vol = 215.636
core size 155 of 198  vol = 206.808
core size 154 of 198  vol = 196.999
core size 153 of 198  vol = 188.554
core size 152 of 198  vol = 182.276
core size 151 of 198  vol = 176.967
core size 150 of 198  vol = 170.725
core size 149 of 198  vol = 166.134
core size 148 of 198  vol = 159.81
core size 147 of 198  vol = 153.78
core size 146 of 198  vol = 149.107
core size 145 of 198  vol = 143.672
core size 144 of 198  vol = 137.152
core size 143 of 198  vol = 132.531
core size 142 of 198  vol = 127.244
core size 141 of 198  vol = 121.587
core size 140 of 198  vol = 116.787
core size 139 of 198  vol = 112.583
core size 138 of 198  vol = 108.182
```

```
core size 137 of 198  vol = 105.145
core size 136 of 198  vol = 101.261
core size 135 of 198  vol = 97.387
core size 134 of 198  vol = 92.986
core size 133 of 198  vol = 88.195
core size 132 of 198  vol = 84.039
core size 131 of 198  vol = 81.908
core size 130 of 198  vol = 78.029
core size 129 of 198  vol = 75.282
core size 128 of 198  vol = 73.063
core size 127 of 198  vol = 70.706
core size 126 of 198  vol = 68.984
core size 125 of 198  vol = 66.715
core size 124 of 198  vol = 64.384
core size 123 of 198  vol = 61.153
core size 122 of 198  vol = 59.037
core size 121 of 198  vol = 56.633
core size 120 of 198  vol = 54.022
core size 119 of 198  vol = 51.805
core size 118 of 198  vol = 49.652
core size 117 of 198  vol = 48.193
core size 116 of 198  vol = 46.648
core size 115 of 198  vol = 44.752
core size 114 of 198  vol = 43.292
core size 113 of 198  vol = 41.093
core size 112 of 198  vol = 39.147
core size 111 of 198  vol = 36.472
core size 110 of 198  vol = 34.117
core size 109 of 198  vol = 31.47
core size 108 of 198  vol = 29.448
core size 107 of 198  vol = 27.325
core size 106 of 198  vol = 25.822
core size 105 of 198  vol = 24.15
core size 104 of 198  vol = 22.648
core size 103 of 198  vol = 21.069
core size 102 of 198  vol = 19.953
core size 101 of 198  vol = 18.3
core size 100 of 198  vol = 15.723
core size 99 of 198  vol = 14.841
core size 98 of 198  vol = 11.646
core size 97 of 198  vol = 9.434
core size 96 of 198  vol = 7.354
core size 95 of 198  vol = 6.179
```

```
 core size 94 of 198  vol = 5.666
 core size 93 of 198  vol = 4.705
 core size 92 of 198  vol = 3.665
 core size 91 of 198  vol = 2.77
 core size 90 of 198  vol = 2.151
 core size 89 of 198  vol = 1.715
 core size 88 of 198  vol = 1.15
 core size 87 of 198  vol = 0.874
 core size 86 of 198  vol = 0.685
 core size 85 of 198  vol = 0.528
 core size 84 of 198  vol = 0.37
 FINISHED: Min vol ( 0.5 ) reached
```

```
core.inds <- print(core, vol=0.5)
```

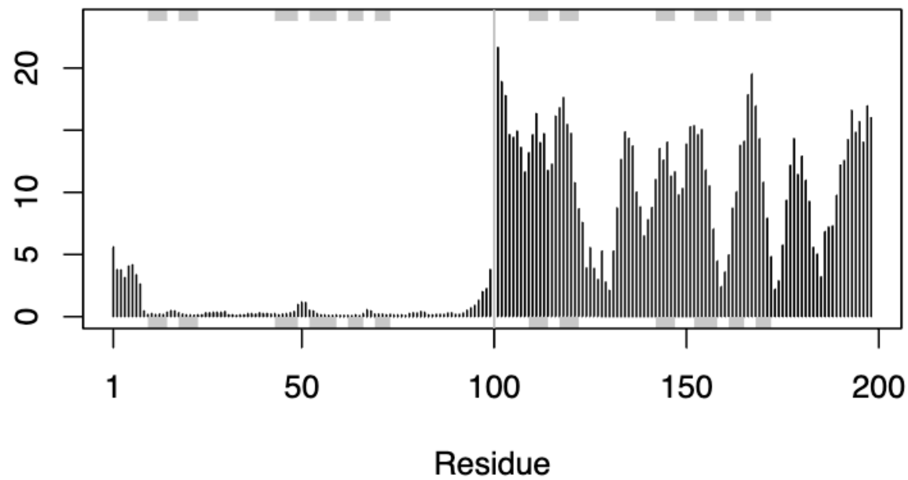
```
# 85 positions (cumulative volume <= 0.5 Angstrom^3)
  start end length
1     9  49     41
2    52  95     44
```

```
xyz <- pdbfit(pdbs, core.inds, outpath="corefit_structures")
```

RMSF Analysis

RMSF measures conformational variance along the structure.

```
rf <- rmsf(xyz)
plotb3(rf, sse=pdb)
abline(v=100, col="gray", ylab="RMSF")
```



Predicted Aligned Error (PAE)

PAE values help us assess confidence in the relative positions of different domains.

```
pae_files <- list.files(path=results_dir,
                        pattern=".*model.*\\.json",
                        full.names = TRUE)

pae1 <- read_json(pae_files[1], simplifyVector = TRUE)
pae5 <- read_json(pae_files[5], simplifyVector = TRUE)

attributes(pae1)
```

```
$names
[1] "plddt"   "max_pae" "pae"     "ptm"     "iptm"
```

```
head(pae1$plddt)
```

```
[1] 90.81 93.25 93.69 92.88 95.25 89.44
```

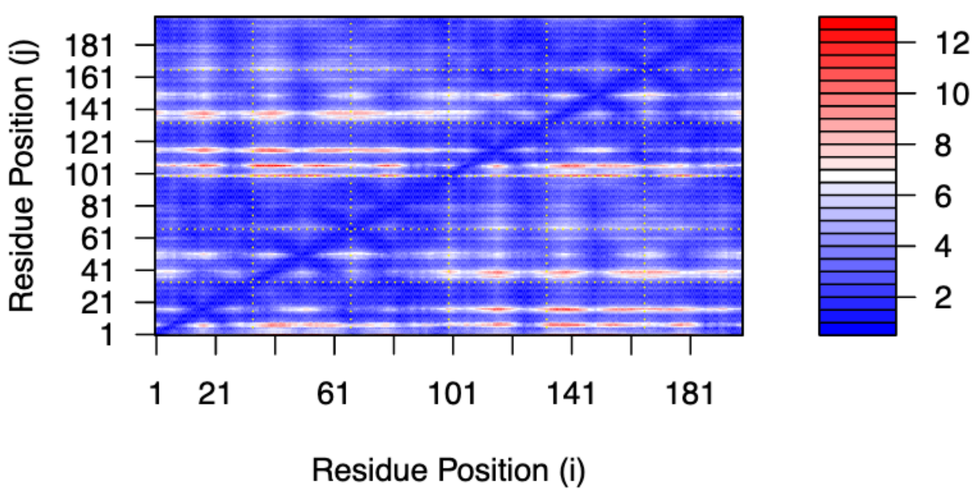
```r
# Compare max PAE scores - lower is better
pae1$max_pae
```

```
[1] 12.84375
```

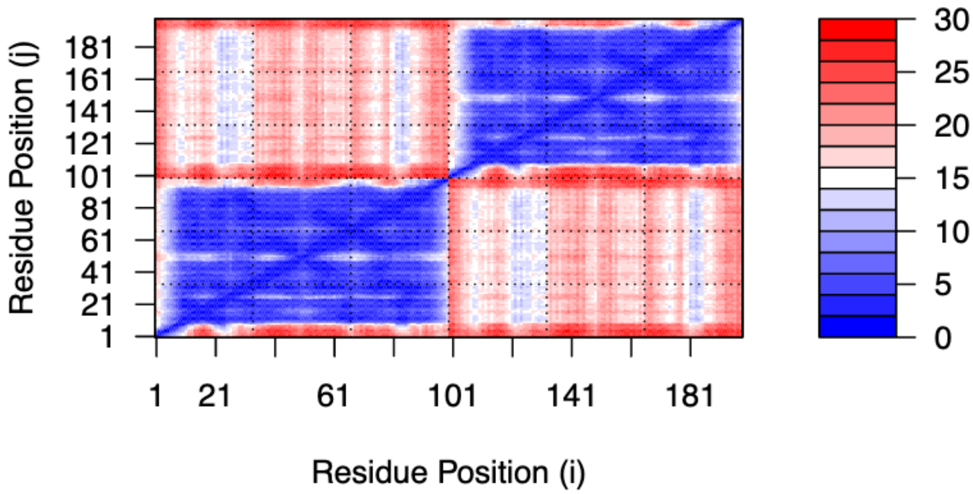
```r
pae5$max_pae
```

```
[1] 29.59375
```

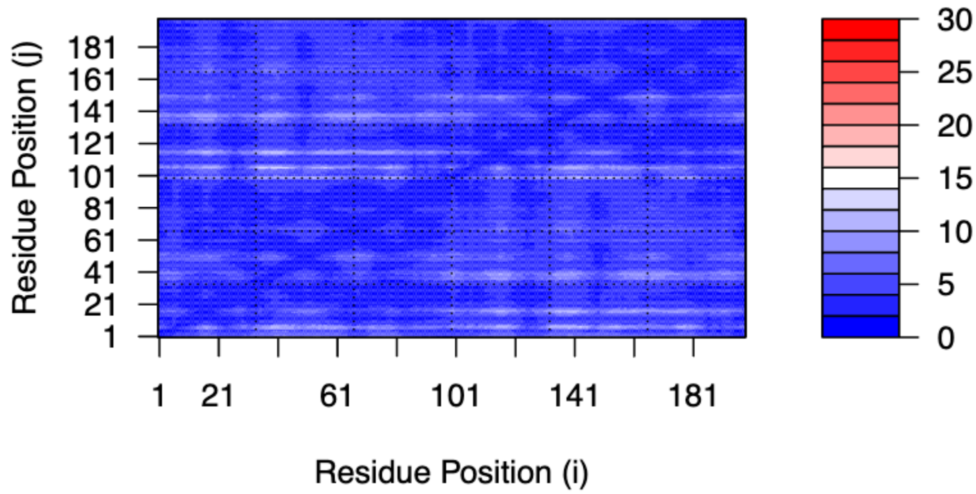
```r
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)")
```




```r
plot.dmat(pae5$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```




```
# Re-plot model 1 with same scale for fair comparison
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```



Residue Conservation Analysis

```
aln_file <- list.files(path=results_dir,
                       pattern=".a3m$",
                       full.names = TRUE)

aln <- read.fasta(aln_file[1], to.upper = TRUE)
```

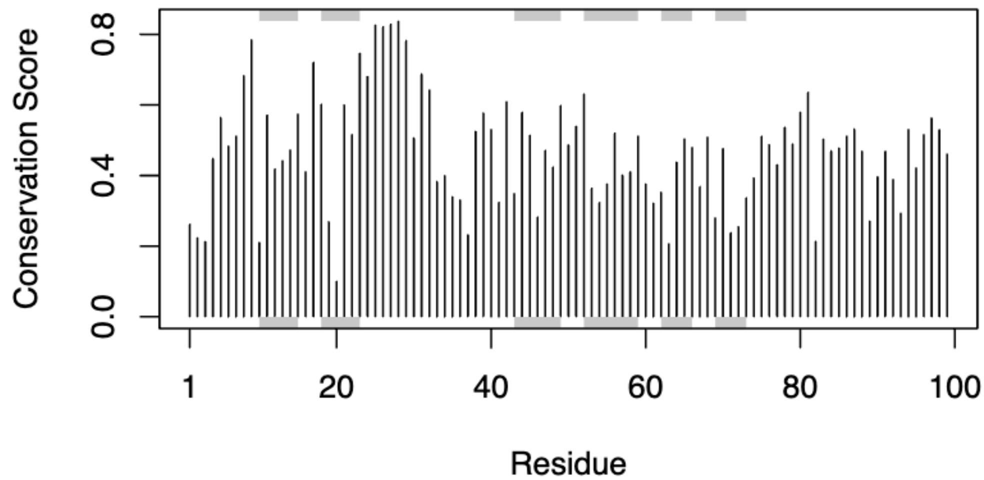
```
[1] " ** Duplicated sequence id's: 101 **"
[2] " ** Duplicated sequence id's: 101 **"
```

```
dim(aln$ali)
```

```
[1] 5397  132
```

```
sim <- conserv(aln)

plotb3(sim[1:99], sse=trim.pdb(pdb, chain="A"),
       ylab="Conservation Score")
```




```
con <- consensus(aln, cutoff = 0.9)
con$seq
```

```
  [1] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [19] "-" "-" "-" "-" "-" "-" "D" "T" "G" "A" "-" "-" "-" "-" "-" "-" "-" "-"
 [37] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [55] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [73] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [91] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[109] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[127] "-" "-" "-" "-" "-" "-"
```

Mapping Conservation to Structure

```
m1.pdb <- read.pdb(pdb_files[1])
occ <- vec2resno(c(sim[1:99], sim[1:99]), m1.pdb$atom$resno)
write.pdb(m1.pdb, o=occ, file="m1_conserv.pdb")
```